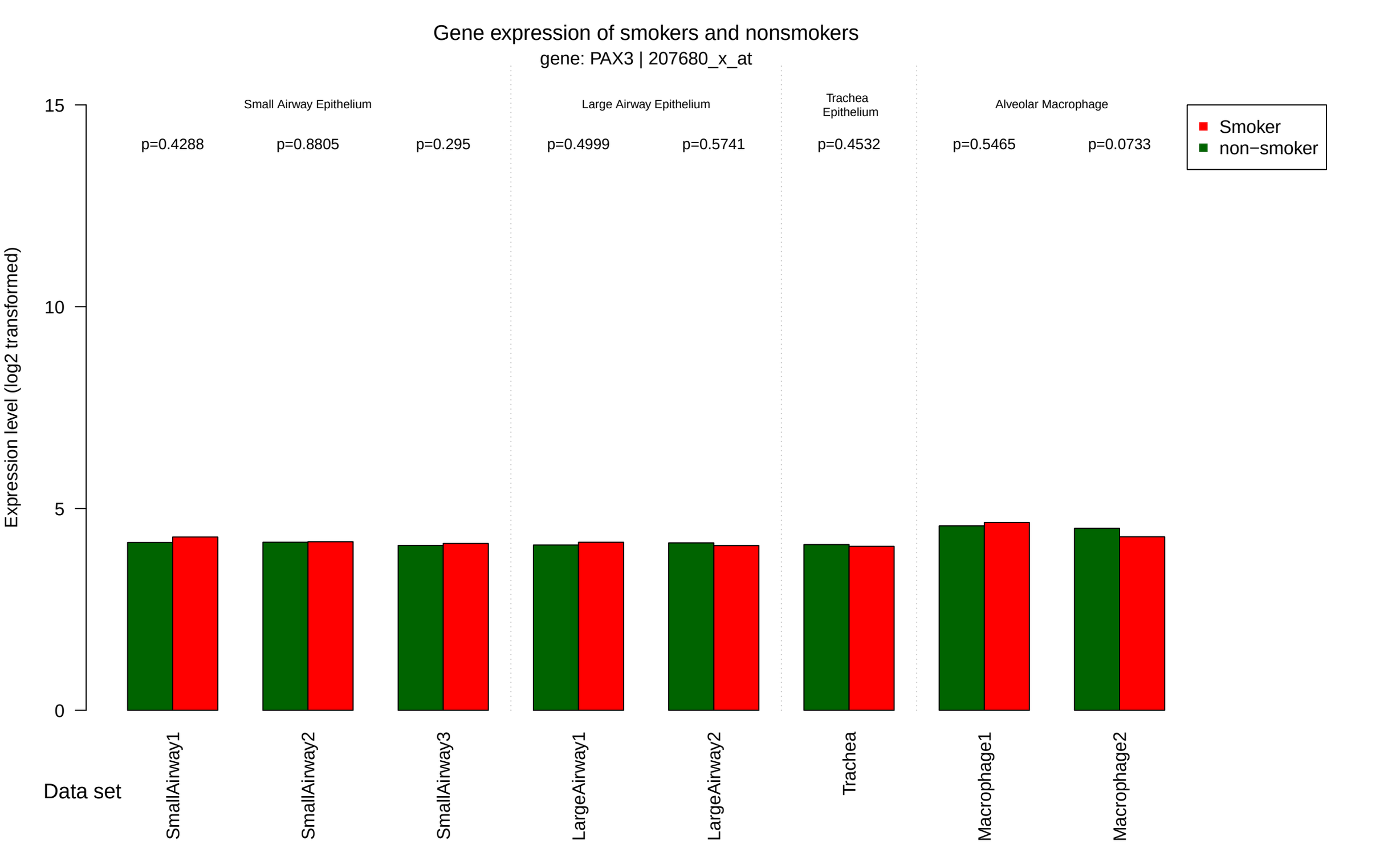
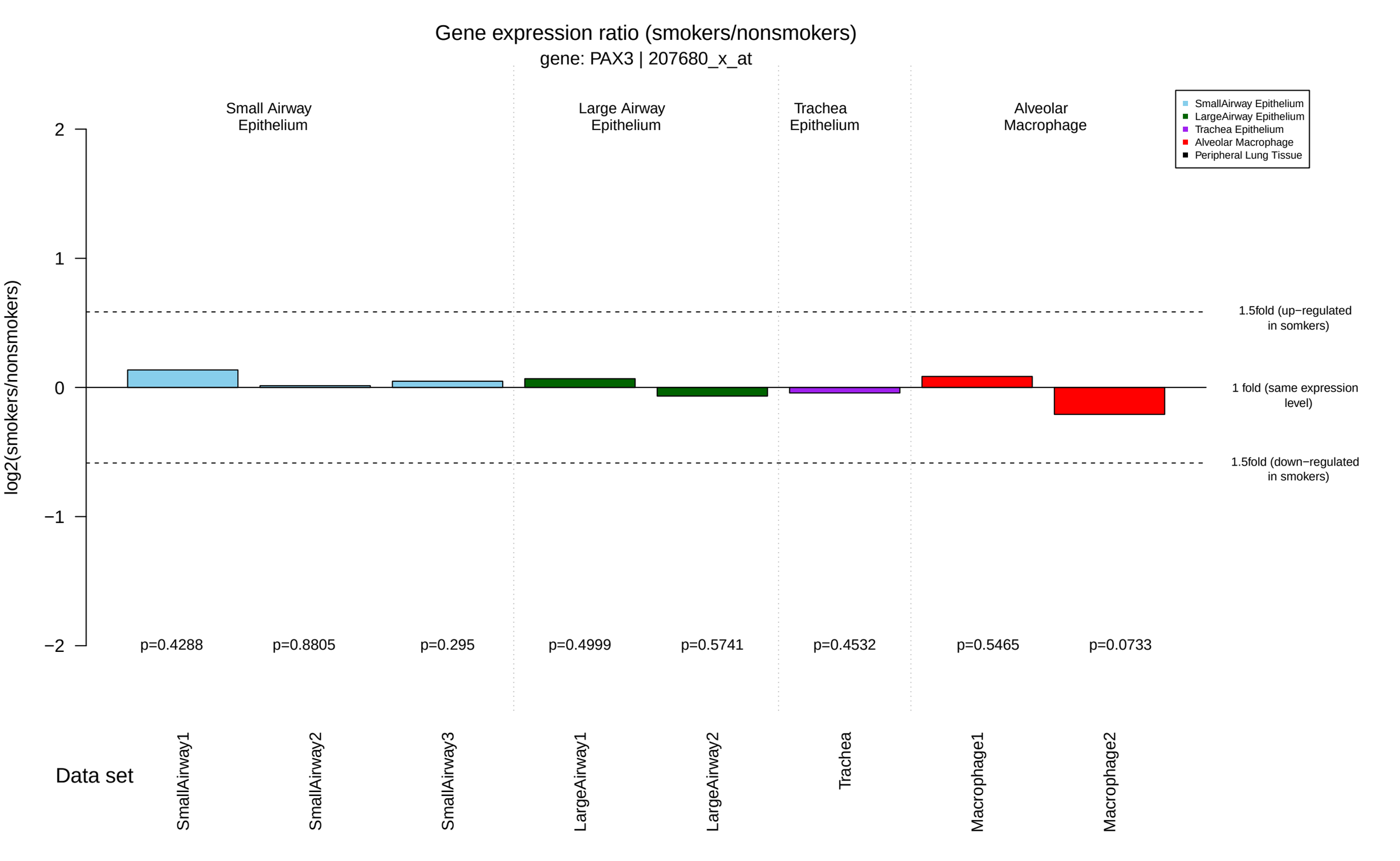
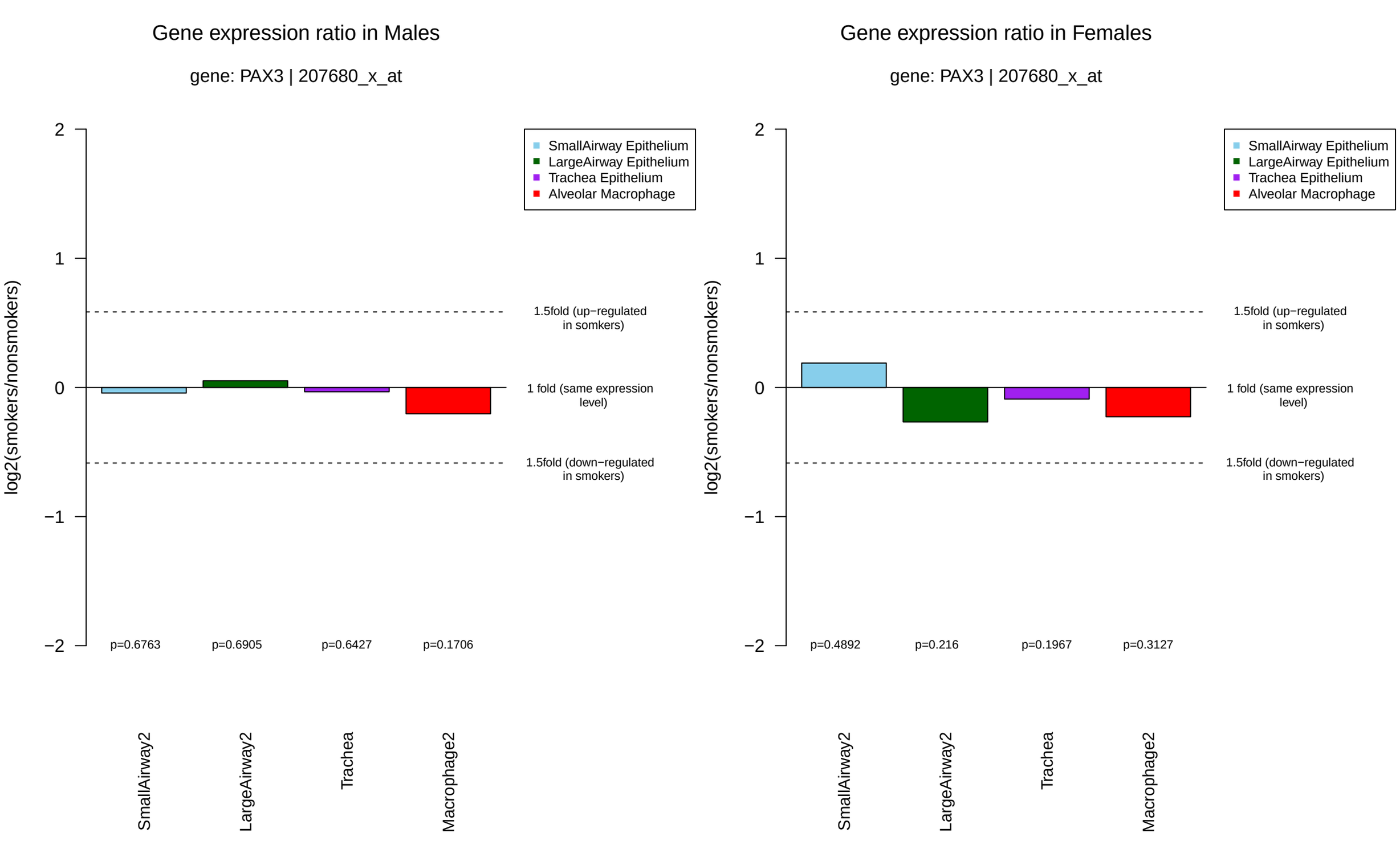
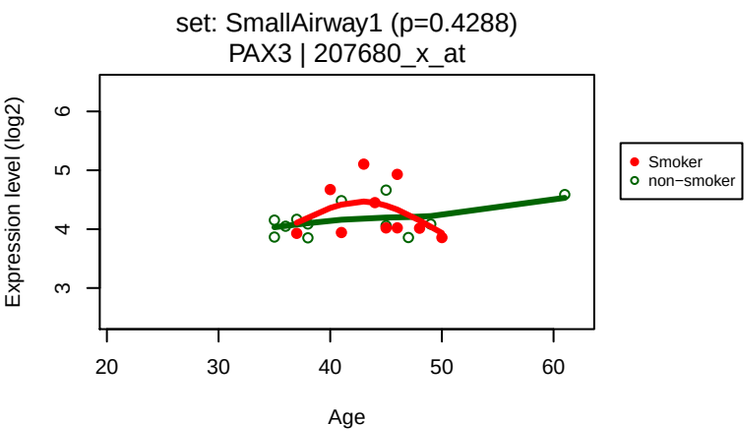
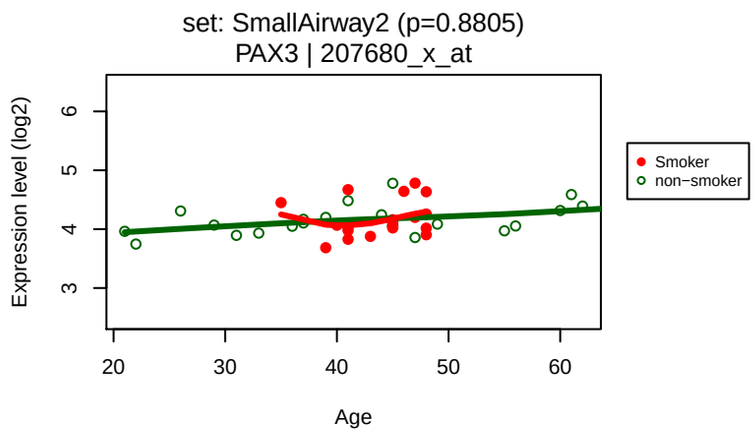
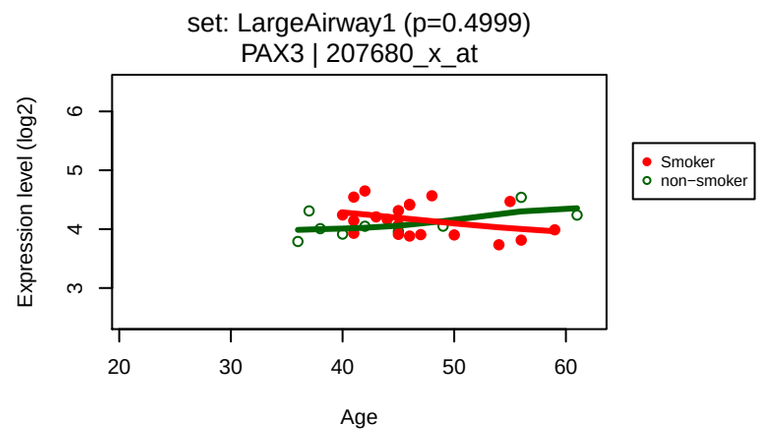

Gene expression of smokers and nonsmokers
gene: PAX3 | 207680_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4288
p=0.8805
p=0.295
p=0.4999
p=0.5741
p=0.4532
p=0.5465
p=0.0733
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

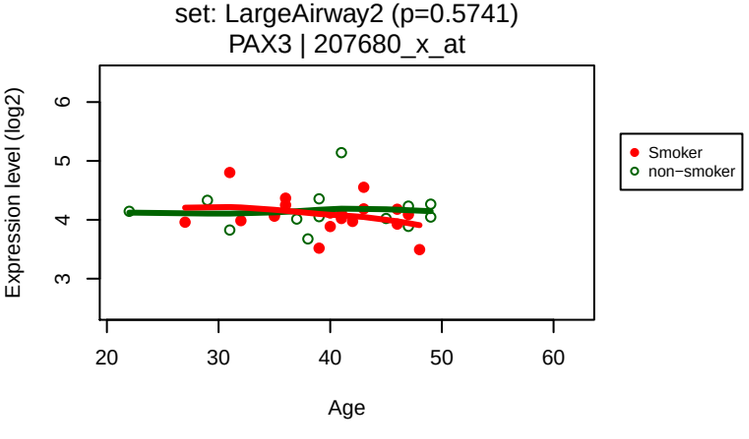

Gene expression ratio (smokers/nonsmokers)
gene: PAX3 | 207680_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4288
p=0.8805
p=0.295
p=0.4999
p=0.5741
p=0.4532
p=0.5465
p=0.0733
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

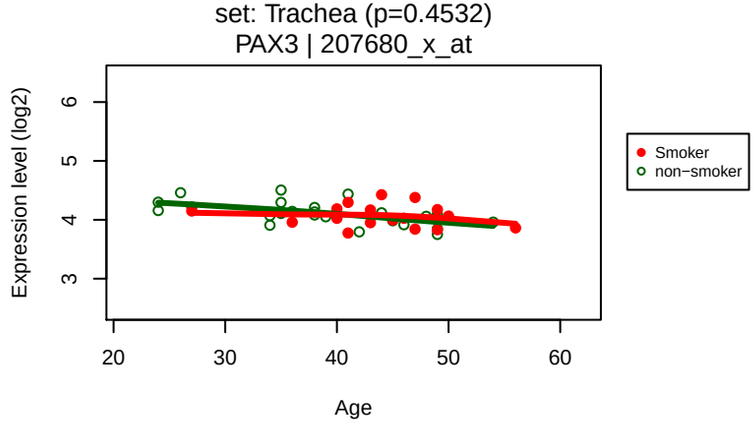

Gene expression ratio in Males
gene: PAX3 | 207680_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6763
p=0.6905
p=0.6427
p=0.1706
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX3 | 207680_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4892
p=0.216
p=0.1967
p=0.3127
SmallAirway2
LargeAirway2
Trachea
Macrophage2

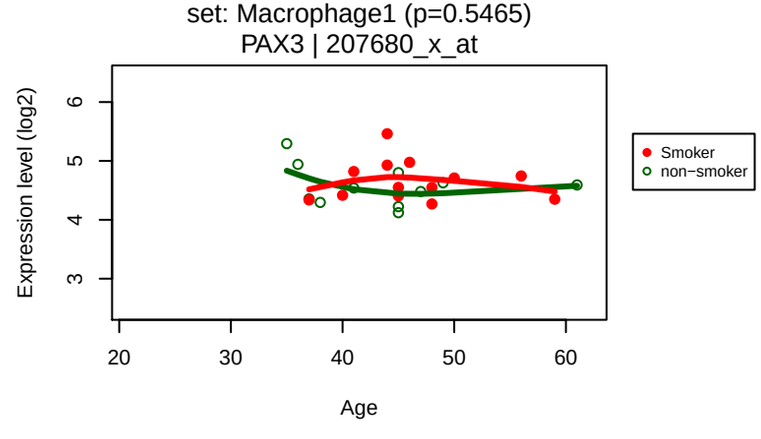

set: SmallAirway1 (p=0.4288)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker

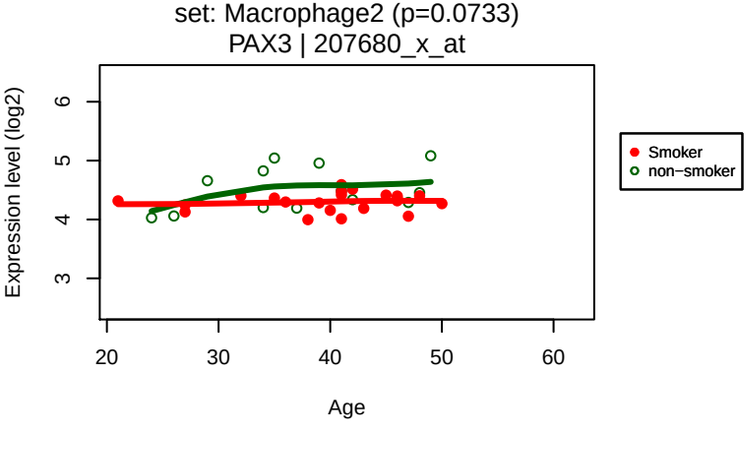

set: SmallAirway2 (p=0.8805)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker

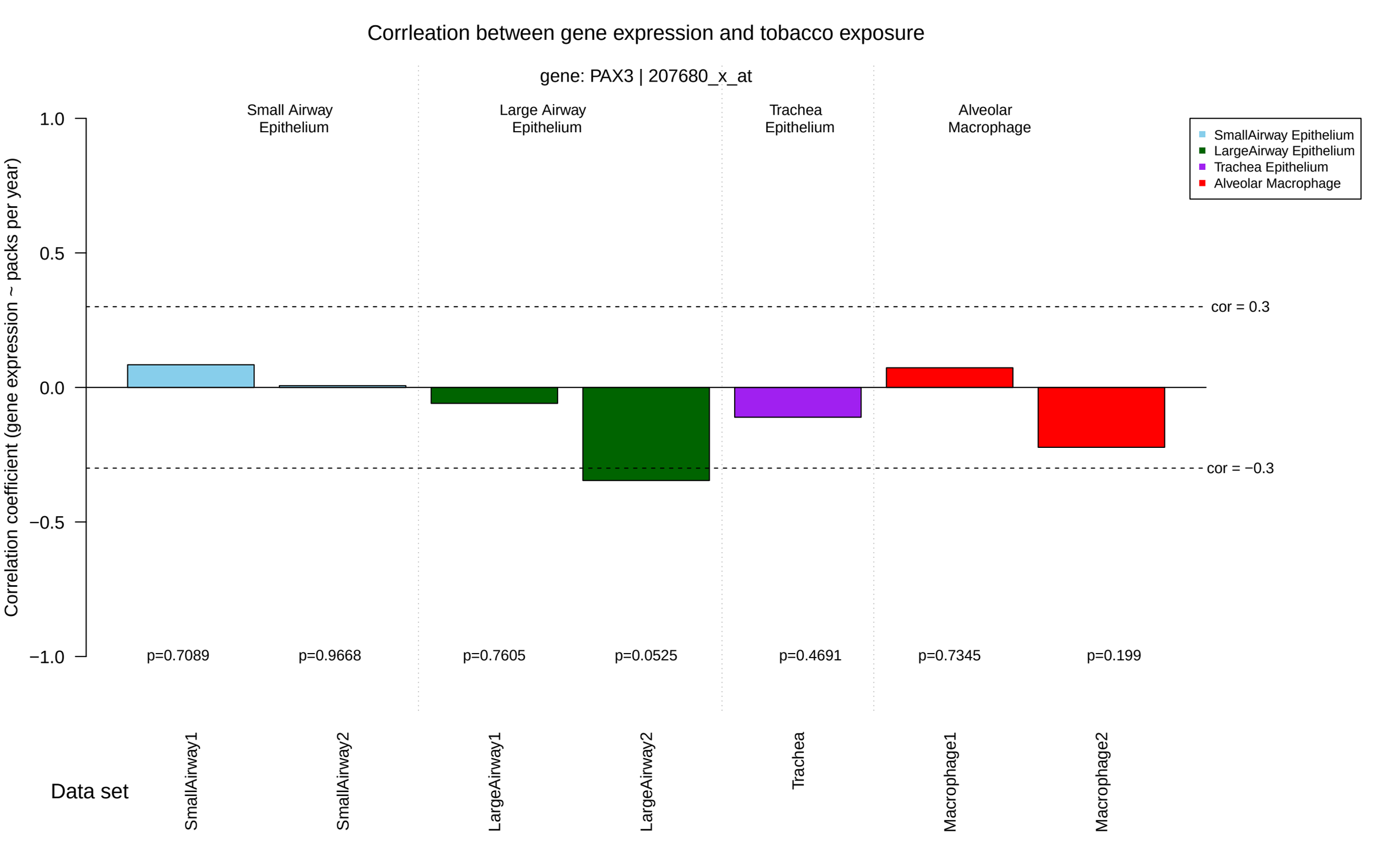

set: LargeAirway1 (p=0.4999)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5741)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4532)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5465)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0733)
PAX3 | 207680_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX3 | 207680_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7089
p=0.9668
p=0.7605
p=0.0525
p=0.4691
p=0.7345
p=0.199
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2